

# CURRICULUM VITAE

Lukas Niebel

September 23, 2026

ETH Zürich

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## Academic appointments

- from 08/2026 **Postdoctoral researcher**, *ETH Zürich*, Mentor: Prof. Dr Mikaela Iacobelli  
04/2023–06/2026 **Postdoctoral researcher**, *University of Münster*, Mentor: Prof. Dr Christian Seis  
04/2019–03/2023 **Research associate**, *Ulm University*

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## Education

- 04/2019–06/2023 **PhD in Mathematics**, *Ulm University*  
Thesis: Analytic aspects of kinetic partial differential equations  
Advisor: Prof. Dr Rico Zacher  
Referees: Prof. Dr Cyril Imbert, Prof. Dr Hongjie Dong  
Grade: Summa cum laude; awarded by the Ulmer Universitätsgesellschaft
- 03/2018–03/2019 **MSc in Mathematics**, *Ulm University*  
Thesis: Kolmogorov equations – Well-posedness, regularity, asymptotics and Harnack inequalities  
Grade: 1.0 with distinction; received the award for the best graduating student
- 10/2014–02/2018 **BSc in Mathematics**, *Ulm University*  
Thesis: Long-time behaviour of Markov chains by discrete functional inequalities and entropic Ricci curvature  
Grade: 1.1 with distinction; received the award for the best graduating student
- 04/2013–09/2014 **Junior studies in mathematics**, *Fernuniversität Hagen and PFH Göttingen*  
06/2014 **Abitur**, *Joachim-Hahn-Gymnasium Blaubeuren*

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## Awards and Scholarships

- 06/2026 DFG Walter Benjamin Fellowship for a research project at the University of Cambridge, hosted by Prof. Clément Mouhot (ca. 100,000 EUR)
- 02/2025 PhD thesis award “Promotionspreis der Ulmer Universitätsgesellschaft” (1500 EUR)
- 06/2022 Travel grant from the ProTrainU Mobility-Programme, Ulm University (1000 EUR)
- 04/2022 Master’s graduation award “Absolventenpreis M.Sc. Mathematik Uni Ulm 2020” for the best graduation (500 EUR)
- 11/2019 Graduate scholarship granted by the State of Baden-Wuerttemberg, Germany (52,200 EUR)
- 05/2018 Bachelor’s graduation award “Absolventenpreis B.Sc. Mathematik Uni Ulm 2018” for the best graduation (500 EUR)
- 06/2014 Ferry Porsche award and graduation award of the Deutsche Physikalische Gesellschaft

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## Research

### Preprints

- [1] **On Leonardo da Vinci’s paradox** with Björn Gebhard and Christian Seis. arXiv:2609.26283, 2026.
- [2] **A Liouville theorem for the two-dimensional stationary hypodissipative Navier–Stokes system** with Nicola De Nitti and Jiaqi Yang. arXiv:2608.16437, 2026.
- [3] **Lions’ maximal regularity problem for divergence-form differential operators: failure at the  $\frac{1}{2}$ -Hölder endpoint.** arXiv:2608.11194, 2026.
- [4] **A fast dynamo with zero topological entropy.** arXiv:2608.02586, 2026.
- [5] **Sharp kinetic trace theory** with Lisa Valentini. arXiv:2607.24708, 2026.
- [6] **On the kinetic  $p$ -Laplace equation with nonlocal diffusion.** arXiv:2605.21080, 2026.
- [7] **Nonlinear kinetic diffusion equations with  $p$ -growth** with Helge Dietert and Rico Zacher. arXiv:2605.18521, 2026.

- [8] **Transfer of regularity by kinetic mollification along critical trajectories.** arXiv:2605.13582, 2026.
- [9] **Spherical rigidity for an exterior overdetermined problem with Neumann data prescribed by mean curvature.** arXiv:2604.07002, 2026.
- [10] **Kinetic Sobolev Spaces** with Pascal Auscher. arXiv:2603.17491, 2026.
- [11] **Nash’s  $G$  bound for the Kolmogorov equation** with Helge Dietert. arXiv:2510.21621, 2025.
- [12] **Global rigidity of two-dimensional bubbles.** arXiv:2510.17557, 2025.
- [13] **Critical trajectories in kinetic geometry** with Helge Dietert, Clément Mouhot, and Rico Zacher. arXiv:2508.14868, 2025.
- [14] **Weak solutions to Kolmogorov–Fokker–Planck equations: regularity, existence and uniqueness** with Pascal Auscher and Cyril Imbert. arXiv:2403.17464, 2024.

#### Publications

- [15] **Steady bubbles and drops in inviscid fluids** with David Meyer and Christian Seis. *Calc. Var.*, 64(299), 2025.
- [16] **A trajectorial interpretation of Moser’s proof of the Harnack inequality** with Rico Zacher. *Ann. Sc. Norm. Super. Pisa Cl. Sci. (5)*, 26(2):605–617, 2025.
- [17] **Fundamental solutions to Kolmogorov–Fokker–Planck equations with rough coefficients: existence, uniqueness, upper estimates** with Pascal Auscher and Cyril Imbert. *SIAM J. Math. Anal.*, 57(2):2114–2137, 2025.
- [18] **On a kinetic Poincaré inequality and beyond** with Rico Zacher. *J. Funct. Anal.*, 289(1):Paper No. 110899, 18, 2025.
- [19] **Analytic aspects of kinetic partial differential equations.** *OPARU — Open Access Repository der Universität Ulm*, 2023.
- [20] **Kinetic maximal  $L^p$ -regularity with temporal weights and application to quasilinear kinetic diffusion equations** with Rico Zacher. *J. Differential Equations*, 307:29–82, 2022.
- [21] **Kinetic maximal  $L^p_\mu(L^p)$ -regularity for the fractional Kolmogorov equation with variable density.** *Nonlinear Anal.*, 214:Paper No. 112517, 21, 2022.
- [22] **Kinetic maximal  $L^2$ -regularity for the (fractional) Kolmogorov equation** with Rico Zacher. *J. Evol. Equ.*, 21(3):3585–3612, 2021.

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#### Employment History

- 10/2022–03/2023 **Predoctoral researcher**, *Ulm University*
- 03/2020–10/2022 **Teaching assistant**, *University of Applied Sciences Neu-Ulm*
- 10/2015–10/2019 **Teaching assistant**, *Ulm University*
- 03/2015–03/2019 **Research assistant (working student)**, *Daimler AG Research & Development Ulm*

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#### Teaching Experience

##### ETH Zürich

- Autumn 2026 Instructor for the student seminar *Collisional Kinetic Theory: Entropy and Fisher Information*
- University of Münster
- 2023–2025 Teaching assistant for *Partial Differential Equations* (1 sem.) and *Analysis and Numerics of Differential Equations* (1 sem.) (exercise-sheet design; large-class solution sessions; tutor and marking coordination; exam support)
- Winter 2024/25 Supervised a student seminar on *De Giorgi–Nash–Moser Theory*
- 2024–2025 Examined two BSc theses; co-supervised and examined the MSc thesis of Yuanjiang Han (now a PhD student with Christian Seis)

## University of Applied Sciences Neu-Ulm

2020–2022 Teaching assistant for *Business Mathematics and Statistics I* (3 sem.) and *Business Mathematics and Statistics II – Regression Models / Statistical Testing* (2 sem.) (authored exercise sheets; ran the programming lab; marked submissions; created teaching videos)

## Ulm University

2017–2022 Teaching assistant for *Analysis I* (2 sem.), *Analysis II* (1 sem.), *Dynamical Systems* (2 sem.), and *Elements of Functional Analysis* (1 sem.) (exercise-sheet design; large-class solution sessions; tutor and marking coordination; exam-setting support)

Winter 2022/23 Local coordinator of the 26th Internetseminar

Winter 2019/20 Organised the “Trainingscamp”, an annual four-week maths skills refresher for first-year students

2015–2017 Matlab tutor for *Analysis I* and *Analysis II*

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## Talks and Posters

- 08/2026 Talk – Nonlinear PDEs, Harmonic Analysis and Free Boundary Problems, KTH Stockholm
- 05/2026 Talk – Calderón–Zygmund Analysis Seminar, University of Chicago
- 05/2026 Talk – Seminar Applied and Computational Mathematics, Tulane University
- 04/2026 Talk – ISU Analysis and Probability Seminar, Iowa State University
- 03/2026 Talk – Horizons in nonlinear PDEs, Ulm University
- 10/2025 Talk – Oberseminar Angewandte Analysis, Dortmund University
- 10/2025 Talk – Noko25 43rd Northern German Colloquium, Humboldt-Universität zu Berlin
- 09/2025 Poster – Hyperbolic Problems in Nürnberg 2025, FAU Erlangen
- 05/2025 Talk – Seminar Nichtlineare Partielle Differentialgleichungen, KIT Karlsruhe
- 03/2025 Poster – Kinetic equations and turbulence, CentraleSupélec and Collège de France
- 10/2024 Talk – Münster–Imperial Day in PDE, University of Münster
- 09/2024 Talk – Oberseminar Analysis, University of Bonn
- 03/2024 Poster – Mathematics Münster Mid-term Conference, University of Münster
- 11/2023 Talk – Séminaire Analyse Numérique et EDP, Université Paris-Saclay
- 09/2023 Talk – Seminar of the AG Analysis-Probability, MPI Leipzig
- 06/2023 Talk – Real Analysis and Geometry, CIRM Marseille
- 04/2023 Talk – Wilhelm Killing Colloquium, University of Münster
- 03/2023 Talk – Chair Dynamics Control and Numerics, FAU Erlangen
- 02/2023 Talk – TULKKA, KIT Karlsruhe
- 11/2022 Poster – Kinetic Theory, CIRM Marseille
- 09/2022 Talk & Poster – Horizons in non-linear PDEs, Ulm University
- 07/2022 Poster – When Kinetic Theory meets Fluid Mechanics, ETH Zürich
- 06/2022 Poster – 11th Summer school on “Methods and Models of Kinetic Theory”, Pesaro
- 01/2022 Poster – Frontiers in analysis of kinetic equations, INI Cambridge
- 07/2021 Talk – Nonlinear Evolution Equations and Approximations, University of Duisburg-Essen
- 03/2021 Talk – Oberseminar Analysis, Ulm University
- 02/2021 Poster – Winterschool on Analysis and Applied Mathematics, University of Münster

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## Research Stays

- 05/2026 Visit to Luis Silvestre (University of Chicago), one week
- 04/2026 Visit to Sam Punshon-Smith (Tulane University), one week
- 04/2026 Visit to David Herzog (Iowa State University), two weeks
- 2023–2025 Multiple visits to Pascal Auscher (Université Paris-Saclay), totalling 7 weeks
- 2023–2025 Multiple visits to Rico Zacher (Ulm University), totalling 5 weeks
- 2021–2025 Three visits to Cyril Imbert (École normale supérieure, Paris), totalling two weeks
- 05/2025 Visit to Björn de Rijk (KIT Karlsruhe), one week
- 09/2023 Visit to Tobias Ried (MPI Leipzig), one week

- 03/2023 Visit to Nicola De Nitti (FAU Erlangen), one week  
 02/2023 Visit to Christian Seis (University of Münster), half a week

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## Other Scientific Activities

- 2024 & 2026 Supporting the organisation of the Münster–Imperial Day in PDE, University of Münster  
 2022 Supporting the organisation of the summer school “Horizons in non-linear PDEs” at Ulm University  
 2020 Co-organisation of the seminar “Mathematisches Kolloquium” at Ulm University for six semesters  
 2022 Organisation of a weekend workshop for mathematics PhD students at Ulm University: 15 participants, secured 2500 EUR in funding from StuVe Ulm  
 2022 Supervision of a three-month project for the workshop of the 25th Internetseminar. Title: Kolmogorov equation in  $L^p(\mathbb{R}^{2n})$  – Semigroup, spectrum, maximal regularity?  
 2021 Facilitator at the autumn school “COLLAB – Collaboratory for Global Sustainability 2021”, Toronto/Ulm  
 2019 Supporting the organisation of the winter school “Grad. Flows and Var. Methods in PDEs” at Ulm University  
 Active contribution to the PDE seminars at the University of Münster and Ulm University by inviting several external speakers  
 Attended over 30 international workshops and conferences, including events at CIRM Marseille, INI Cambridge, ETH Zürich, and the University of Oxford  
 Peer reviewer for *Annales scientifiques de l'École normale supérieure*, *Communications in Partial Differential Equations*, *Discrete and Continuous Dynamical Systems*, *European Journal of Mathematics*, *Journal of Evolution Equations*, *Mathematika*, *Nonlinearity* and *Potential Analysis*

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## Academic Volunteer Work

- 02/2021–09/2022 **Elected member of the council of PhD students**, *Ulm University*  
 Representing the interests of doctoral students in university committees, e.g. for the change of the PhD regulations; chair of the council of PhD students (12/2021–09/2022)  
 02/2021–09/2022 **Elected member of the faculty council for Mathematics and Economic Sciences**, *Ulm University*  
 07/2017–03/2019 **Member of the Mathematics Student Council**, *Ulm University*  
 Member of the appointment committee for a professorship in “Stochastics”